POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2025
Programme: BE		Full Marks : 100
Course: Artificial Intelligence (New)		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) If computer passes Turing test, will it act like human? justify with reasons. What are impacts of AI in society? 7

   b) What is PEAS? Explain PEAS representation for an autonomous taxi agent. 8

2. a) Why is searching important in problem solving? What are the drawbacks of Greedy Best First Search and how A* search technique is used to solve it. Explain with example. 7

   OR

   Why is Hill Climbing called as Local greedy search? Explain the problems in Hill Climbing with appropriate diagram.

   b) what are the constraint satisfaction problems? Solve the following crypt arithmetic problem **BASE+BALL=GAMES** 8

3. a) Explain any four knowledge representation techniques used in AI. 8

   b) Consider the following axioms: 7

      i. Ravi likes all kind of food.
      ii. Apples and chicken are food
      iii. Anything anyone eats and is not killed is food
      iv. Ajay eats peanuts and is still alive
      v. Rita eats everything that Ajay eats.

   Prove by resolution that Ravi likes peanuts using resolution.

4. a) Explain Bayesian Networks with example. 8

   b) What do you understand by Learning by analogy? Explain in detail. 7

5. a) How do you differentiate K means clustering and K nearest neighbor algorithm? Using K-means Clustering, cluster the following data into clusters. Where k= 32 for data: 8

	A	B
1	2	5
2	3	6
3	5	4
4	7	7
5	3	5
6	5	3

b) Differentiate between Supervised Learning and Unsupervised Learning with example. 7

6. a) What do you mean by membership function in fuzzy logic? Differentiate two types of membership functions: Triangular and Trapezoidal method. 8

OR

Difference between Fuzzification and Defuzzification. Explain in detail.

b) Explain the architecture of expert systems in detail. 7

7. Write short notes on: **(Any two)** 2×5

a) Alpha-beta pruning

b) Semantic Net and Frames

c) Forward and Backward Chaining

Level: Bachelor
Programme: BE
Course: Engineering Management

POKHARA UNIVERSITY

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define management and explain the major functions of management. 7

   b) List out the importance of management in technology. What are the roles and responsibilities of an engineering manager? 8

2. a) Define planning. Explain the levels of planning. 8

   b) Compare traditional structure with modern structure of organization. List out some of the emerging issues while organizing ICT companies. 7

3. a) Do you think Maslow's hierarchy is still relevant theory at present context? Explain some of the techniques of motivation. 8

   b) Discuss the challenges in motivating and leading a technical workforce and suggest strategies to overcome them. 7

4. a) Explain the types of control. Give some examples of ICT tools for effective control of engineering projects. 8

   OR

   Why are on the job training necessary? Explain the benefits of it.

   b) Define human resource management and bring out the need of HRM in an organization. Explain six HRM functions. 7

5. a) Define interpersonal conflict. What would you do to resolve it? Explain. 8

   OR

   What is your opinion regarding conflict in organization? Are conflicts always bad for organizations? What are the sources of conflict?

   b) What are the emerging trends in engineering management? Describe any two with their key benefits. 7

CS CamScanner

6. a) Human resource management is a complex process compared to other resources. Justify. 7

b) What is meant by learning organization? List out the characteristics of a learning organization in the ICT industry. 8

7. Write short notes on: **(Any two)** 2×5

a) Levels of Management

b) Hybrid organization

c) Performance appraisal

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Embedded System (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate embedded systems and general-purpose systems based on architecture, performance, and applications. 7
   b) What is the utility of PORTx, PINx, and DDRx registers in AVR? Write a program using all these three kinds of register. Explain what that program can do. 8
2. a) Explain the AVR microcontroller architecture with a neat block diagram. Describe the role of registers, memory, and I/O ports. 7

   OR

   Explain interrupt handling mechanism in AVR microcontrollers.
   b) Explain the conditions that can lead to deadlock in RTOS. 8
3. a) Explain context switching in RTOS with a diagram. Can an embedded system function reliably without an RTOS? Justify. 7
   b) Explain different modeling styles in VHDL. Write VHDL code for a 2:1 multiplexer using behavioral modeling. 8
4. a) Write a VHDL code for a full adder using behavioral modeling. Explain the data types used and draw the simulation waveform. 7
   b) Write a VHDL code to detect a sequence '1010'. 8
5. a) A company is designing a smart home automation system in which various sensors and actuators communicate with a central hub. The system should support wired or wireless communication for different devices. 7
      i. Identify and justify suitable communication protocols (wired/wireless) for this system.
      ii. Compare SPI, I2C, and UART in terms of data rate, complexity, and power consumption in this context.
   b) Write an AVR C program to display the message 'Hello AVR' on a 16 by 2 LCD in 4-bit mode. 8

6. a) A stepper motor is used in a robotic positioning system. 7

   i. Explain the working principle of a stepper motor and its advantages in embedded systems for precise control.

   ii. If the stepper motor requires 200 steps for a complete revolution. Calculate the step angle of the motor.

   iii. For the same motor, compute the number of steps required to rotate it by 90°?

   b) For a smart home automation system; 8

   i. Explain the role of Arduino.

   ii. List and explain sensors & actuators used.

   iii. How does Arduino support communication between different devices?

   iv. State limitations of Arduino for large-scale automation systems.

OR

What is an MQTT client? Write the steps involved in establishing an MQTT connection between an IoT device and a broker.

7. Write short notes on: (**Any two**) 2×5

   a) Components of an embedded system

   b) TCP/IP

   c) ESP32

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Signal Analysis and Processing (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary code words representing samples of an input signal The link is operated at 10,000 bits/sec and each input sample is quantized into 512 different voltage levels. 8

   i. What is the sampling frequency and folding frequency?

   ii. What is the Nyquist rate for the signal?

   iii. What are the frequencies in resulting discrete time signal x(n)?

   iv. What is the resolution?

   b) Determine the convolution sum of two sequences using graphical method: 7

$$x(n) = \{4,2,1,3\}, h(n) = \{1,2,2,1\}$$

2. a) Determine the z-transform and ROC of $x(n) = 5^n u(n)$ and $x(n) = \cos(wn)u(n)$. 8

OR

Define Z transform and explain the properties of ROC of Z transform.

   b) An LTI system is described by the equation 7

$$y(n) = x(n) + 0.81x(n-1) - 0.81x(n-2) - 0.45y(n-2)$$

Determine the transfer function of the system. Sketch the poles and zeros on the z-plane. Assess the stability.

3. a) Realize the following system by its direct form I and form II structure. 7

$$y[n] = x[n] - 2x[n-1] + 0.5x[n-2] + 0.8y[n-2]$$

b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure. 8

$$H(Z)=1+\frac{13}{24}z^{-1}+\frac{5}{8}z^{-2}+\frac{1}{3}z^{-3}$$

4. a) Determine H(z) using Impulse invariant technique for the analog system function. 7

$$H_a(s)=\frac{1}{(s+0.5)(s^2+0.5s+2)}$$

b) Design a low pass Butterworth digital Filter to give response of 3dB or less for frequencies upto 2kHz and attenuation of 20dB or more beyond 3kHz. Use the Bilinear transformation technique and obtain H(z) of the desired filter. Take sampling frequency as 8kHz. 8

OR

Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.9 \le |H(\omega)| \le 1;\ 0 \le \omega \le \frac{\pi}{2}$$

$$|H(\omega)| \le 0.2;\ \frac{3\pi}{4} \le \omega \le \pi$$

with $T = 1$ s.

5. a) How FIR digital filters are designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7

b) Design a linear FIR filter using Kaiser window to meet the following specifications: 8

$$0.99 \le | H(e^{jw} | \le 1.01;\ \text{for } 0 \le |w| \le 0.19\pi$$

$$| H(e^{jw} | \le 0.01;\ \text{for } 0.21\pi \le |w| \le \pi$$

6. a) Find the 8-point DITFFT of x(n) = {2, 2, 2, 2, 1, 1, 1, 1}. 8

b) Find the circular convolution of $x_1(n) = \{1,2,1,2\}$ and $x_2(n) = \{4,3,2,1\}$ by the graphical method. 7

7. Write short notes on: (**Any two**) 2×5

a) Properties of system

b) Gibbs phenomena and its remedy

c) Chebyshev Vs elliptic filter

Level: Bachelor
Programme: BE
Course: Probability and Statistics (New)

POKHARA UNIVERSITY

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) A company monitors the download times in seconds of two software versions, version X and version Y, over 50 trials each. 7

Download time	5-7	7-9	9-11	11-13	13-15	15-17	17-18
Version X	5	10	12	10	8	3	2
Version Y	2	4	5	10	12	10	7

Which version has greater uniformity?

b) A tech company deploys updates to its software through three servers: Server X, Server Y and Server Z. Server X handle 50% of all updates, server Y handles 30% and server Z handles 20%. Historically, the probability that an update fails on server X is 2%, on server Y is 3% and on server Z is 5%. An update is randomly selected and found to have failed. What is the probability that this update came from server X, from server Y and from server Z? 8

2. a) A random variable X has the following probability function: 7

X	-2	-1	0	1	2
P(x)	0.2	0.1	0.3	0.3	0.1

i. $E(x)$
ii. $E(2x-3)$
iii. $V(2x-3)$

b) Out of 800 families in a certain city with 4 children each, what percentage would be expected to 8
i. 1 boy and 3 girl.
ii. at least one boy find the Expected number of families having 2 boys. Assume boys and girl having equal probabilities.

3. a) The mileage in thousands of miles which car owners get with a certain kind of type is a r.v having exponential distribution with 7

$\lambda = 0.5$. What is the probability that such bulb will be still in operating condition after 4000 hours?

OR

In an examination 15% of the student got first class 60 marks or above, while 40% securing below 40 marks. Assuming the marks are normally distributed, estimate the mean and standard deviation.

b) An office switchboard receives telephone calls at the rate of 3 calls per minute on an average. If receiving of calls follows a Poisson distribution, find the probability of receiving: 8

i. no calls in one minute interval

ii. at least 3 call in a one minute interval

iii. at most 2 calls in 5 minute interval

4. a) Let the joint probability density function of two continuous random variables X and Y be given by 7

$$f(x, y) = f(x) = \begin{cases} k(x+y), & 0 < x < 2, 0 < y < 1 \\ 0, & elsewhere \end{cases}$$

i. Find the value of constant k.

ii. Obtain the marginal probability density function of X and Y.

iii. Find $P(X < 1, Y < 0.5)$

iv. Determine whether X and Y are independent or not.

b) A software company wants to estimate the average time in minutes required by its IT support team to resolve a specific type of software bug. A random sample of 8 bug-resolution time in minutes is recorded as follows: 8

42	38	45	40	44	39	41	43

Assuming that the bug-resolution times are approximately normally distributed, construct a 95% confidence interval for the population mean resolution time using t-distribution.

5. a) A population consists of the value 1, 2, 3, 6, 8. Prove that sample mean of size 3 is unbiased estimate of population mean (without replacement). 7

OR

Two independent random samples of 70 boys and 55 girls from normal population were drawn and found their mean weights were 62 kg and 46 kg with variances 16 and 12 respectively. Construct a 95% confidence for difference between mean weight of boys and girls.

b) Random sample of 400 male and 600 female were asked whether they would like to have a airport near their residence. 200 men and 325 women were in favor of the proposal. Test the hypothesis that proportion of men and women are in favors of proposal at 5% level of significance. 8

6. a) A dietitian wishes to see if a person's cholesterol level will change if the diet is supplemented by a certain mineral. Six candidates were pretested and then took the mineral supplement for a sis week period. The results are shown in the table. Can it be considered that the cholesterol level has been changed at the 0.1 level of significance? 8

Individuals	1	2	3	4	5	6
Before	210	235	208	190	172	244
After	190	170	210	188	173	228

b) General Manager of the New Laptop Shop of Tamrakar is concerned about the sales behavior of Lenovo brand laptops sold at the shop. He realizes that many factors might help the sales, among price major ones. Following data recorded. 7

y (Sales unit)	50	40	70	80	60	55
x (Price in '000 Rs)	70	50	80	90	100	45

i. Estimate the simple regression equation that best fits the data.

ii. Predict the value of sales when the price is Rs. 75000.

iii. Find coefficient of determination and interpret.

iv. Find correlation coefficient and interpret.

7. Write short notes on: (**Any two**) 2×5

a) Characteristics of good estimator

b) Error in hypothesis testing

c) Application of statistics in engineering

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Software Engineering (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain object-oriented analysis and its importance in requirements engineering. 7

   b) Compare and Contrast Waterfall and Incremental Model in detail. 8

   OR

   Explain Agile software development in detail. Compare Scrum and Extreme Programming with workflows and roles.

2. a) Explain software reuse approaches including frameworks, product lines, and COTS reuse with benefits and risks. 7

   b) Define risk. How do you identify risk in software engineering and also explain RMMM briefly. 8

3. a) Why is software engineering required? Describe the 4P's of software project management. 7

   b) Explain the Layered, client server and pipe and filter architecture with suitable diagrams. 8

4. a) What do you mean by black box and white box testing? Discuss the basis path testing and Cyclomatic Complexity of white box testing. 7

   b) What is software quality control and software quality assurance? Explain why reviews and inspections is essential in the SQA process. 8

5. a) Define software configuration management. Explain the process of system building and system release with suitable example. 8

   b) Why do we need to follow software process model while developing software projects. Justify above statement by taking one agile software process model highlighting its important aspects. 7

6. a) A customer goes to the restaurant and orders two pizzas and one coffee through restaurant application. Each table in the restaurant has LED display with the application installed. The customer must 8

login to the restaurant application in order to place the order. After 20 minutes of placing the order food is served by a Robot at the custome's table. The customer pays the bills through the application and the system also allows the customer to provide feedback.

From this scenario

- **a. Draw use case diagram**
- **b. Sequence diagram**
- **c. Class diagram**

b) For which type of software projects is the Spiral Model most appropriate? Justify your answer. 7

OR

What do you mean by agile development? Differentiate between Scrum and Extreme Programming.

7. Write short notes on: (**Any two**) 2×5

~~a)~~ Alpha Vs Beta testing

b) Software Re-Engineering

c) Aspect oriented software engineering